

ENCODE CLUB · PROGRAMMABLE MONEY HACKATHON · BUILD ON ARC

Street Rail.

Streetwear + onchain move rights, settled in stablecoins on Circle's Arc.

H2H · H2A · A2A

USDC gas

EURC + clrBTC

Privy · Google login

Verified on Arcscan

THE PROBLEM

Choreographers built the internet's dance layer.

They got paid \$0.

Missing

No registry of who made which move.

\$0

paid to the choreographer behind the most-copied TikTok dance of 2025.

Missing

No settlement rail once it goes viral.

Street dance already knows how to give credit.

We give it a settlement layer.

CULTURE

"Credit or catch a fade."

Crews, battles, and callouts already enforce authorship offline. The rules exist — the receipts don't.

ONCHAIN

```
log(token, amount, cid)
```

One call. Immutable. Priced in USDC / EURC / cirBTC. Arcscan is the receipt.

Two products. One repo. Same wallet.

TRACK · AGENTIC ECONOMY

Rights Registry

Log any dance move onchain in one call. AI agent quotes, pays, and files the receipt autonomously.

TRACK · DEFI

Streetwear Shop

Sneakers, snapbacks, jackets. Checkout in USDC, EURC, or cirBTC via Circle App Kits.

Four interfaces, one rail.

H2H

HUMAN TO HUMAN

The shop you already know.

- Browse, cart, checkout
- Privy Google login
- USDC / EURC / cirBTC pay

H2A

HUMAN TO AGENT

You delegate, the agent buys.

- Spend policy caps
- Confirm above threshold
- Auditable run ledger

A2A

AGENT TO AGENT

Two agents negotiate, no human.

- A2A 0.3 message/send
- AP2 mandates + UCP checkout
- x402 settlement

A2H

AGENT TO HUMAN

The agent starts. You get paid.

- Royalties pushed, not claimed
- Asks when over mandate
- Receipt in the inbox

All four are live in the app behind one toggle — H2H / H2A / A2A / A2H — and settle to the same contract on Arc Testnet, in stablecoins.

Nobody builds agent-to-human, because there was no rail to push value over.

H2H

HUMAN → HUMAN

Shop, cart, checkout. The interface everyone has already built.

H2A

HUMAN → AGENT

Delegate a purchase under a spend mandate. The agent buys.

A2A

AGENT → AGENT

Buyer and seller agents negotiate and settle over x402.

A2H

AGENT → HUMAN

The agent initiates. Royalties are pushed, approvals requested.

In StreetRail the Rights Agent pays the choreographer in USDC the moment a move earns, pauses when the payout breaks the AP2 mandate, and drops an Arcscan receipt in a payout inbox. It also pushes limited-time merch offers that users can claim on-chain — agent-side receipt, no user gas, discount applied at checkout. Every mandate carries an on-chain authorization block; renewing it extends both the Ed25519 mandate and the contract approval.

LIVE ON ARC

Shipped. Deployed. Verified.

5042002

Arc Testnet chain id

0.8.24

solc, verified on Arcscan

6

decimals — USDC is gas

DANCEMOVETOKENS.SOL

0x4d13b45f823f8944522890c20d8695b6005465f0

Move + purchase log · USDC · EURC · cirBTC

STREETRAILAUTHORIZER.SOL

0x0519c703cde7cbff6829fdcdcfe8c9a4c7aac327

ERC-1271 authorization · owner is the treasury wallet

Both verified on Arcscan. Live wallet balances, agent discovery and payout receipts read straight off Arc — nothing in the demo is mocked.

You set the policy. The agent spends inside it.

01

Intent

"Buy a size L snapback under 25 USDC"

02

Policy

Per-item cap, daily cap, confirm threshold.

03

Interrupt

Above the threshold the human approves first.

04

Pay

Agent signs USDC / EURC / cirBTC from treasury.

05

Ledger

Every step logged, Arcscan tx as the receipt.

Autonomous spending in USDC · decision logic in the agent, not the UI.

A standards stack, not a custom flow.

A2A 0.3

Agent card discovery + message/send — inbound for A2A, outbound for A2H

AP2

Intent, cart, and payment mandates the buyer signs

UCP

Discovery, checkout, order + conformance self-test

x402

402 challenge, payment payload, verified receipt

Arc Testnet

USDC settlement, real tx hash on Arcscan

Buyer and seller agents negotiate live in GX mode, then settle on-chain. In A2H the same stack runs outbound: the AP2 mandate is the standing authorization, x402 settles, Arc is the receipt — and the mandate digest is anchored on-chain, so a counterparty can verify it without ever calling StreetRail.

Six Circle products, **one rail.**

Circle Wallets

Developer-controlled treasury EOA on Arc Testnet — the Rights Agent's own wallet

Circle Contracts (SCP)

DanceMoveTokens deployed and verified on Arcscan, USDC-gas deploy, no EOA key

Nanopayments

Gateway batching (@circle-fin/x402-batching) signs EIP-3009 for the A2A x402 loop

App Kits

Unified Balance for spendable USDC, Swap Kit rates behind the USDC/EURC/cirBTC toggle

Gas Station

Sponsors agent gas on Arc. Paymaster is intentionally unused — USDC is already the gas token

Agent Stack

Buyer agent discovers payable services via Circle's x402 Marketplace Discovery API

ERC-1271 authorizer

Treasury-owned contract signs for the agent — Gateway actions need no EOA delegate

Every rail degrades safely: if Gateway batching or a kit is unavailable on testnet, the agent settles with a direct Arc transfer and says so in the receipt.

The treasury signs for the agent — no EOA delegate.

Approved digest

The treasury sends one transaction to approve a mandate digest. `isValidSignature(digest, 0x)` then returns the magic value. No delegate, no exported key.

Delegate signer

Optional: a time-boxed EOA may sign digests on the treasury's behalf. Revocable on-chain, expires by default.

APPROVED DIGEST

0x1626ba7e

UNKNOWN DIGEST

0xffffffff

StreetRailAuthorizer · 0x0519c703cde7cbff6829fdfdcfe8c9a4c7aac327 · verified on Arcscan. Anyone can check a mandate at </api/public/erc1271/authorizer>.

Agent-to-human offers, **logged on Arc.**

01 Offer

Drop Agent pushes a limited-time merch discount to the inbox.

02 Claim

User taps Claim — no wallet prompt, no user gas.

03 Log

Treasury logs srclaim:<id> to DanceMoveTokens via Circle SCP.

04 Receipt

Claim code + Arcscan tx + signed AP2 OfferClaim mandate.

05 Checkout

Discount is applied when the user pays in USDC / EURC / cirBTC.

REGISTRY

0x4d13b45f823f8944522890c20d8695b6005465f0

AUTHORIZER

0x0519c703cde7cbff6829fdfdcfe8c9a4c7aac327

Claims do not count against the payout cap and are audited on Arcscan.

Programmable money for the culture.

MULTI-STABLECOIN CHECKOUT

Shopify catalog priced in USD, settled in USDC, EURC, or cirBTC. App Kits Unified Balance picks the token with headroom.

NANOPAYMENT ROYALTIES

arc-nanopayments streams sub-cent USDC per play back to the choreographer's wallet. Continuous, not periodic.

Every judging bullet has a feature behind it.

Meaningful use of Arc / USDC

Contract deployed + verified, USDC as gas token

Programmable money flows

Autonomous agent settlement + Gateway-batched nanopayments

Interoperable commerce

H2H, H2A, A2A and A2H on A2A 0.3 + AP2 + UCP + x402

Uses App Kits

Unified Balance for spendable USDC, Swap Kit rates behind the token toggle

Agent with decision logic

Live FX + AP2 spend policy pick the token, cap, and payout

Autonomous spending in USDC

Treasury wallet signs via Circle SCP — no manual approval

Uses Agent Stack

Buyer agent resolves services through Circle's x402 Marketplace Discovery

Verifiable authorization

ERC-1271 contract returns 0x1626ba7e for an approved mandate digest

Working today. Shipping through **Demo Day.**

SHIPPED

- ✓ Contract verified on Arcscan
- ✓ Privy Google login + embedded wallet
- ✓ 7 SKUs live in Shopify (dev store)
- ✓ USDC / EURC / cirBTC token switcher
- ✓ ERC-1271 authorizer verified on Arcscan
- ✓ Circle x402 Discovery wired into negotiation
- ✓ Mobile-tuned end to end
- ✓ On-chain claim offers (A2H)
- ✓ Treasury balance panel + low-gas warning

NEXT · AUG 9 · AUG 20

- Encode Club Demo Day — 9 Aug 2026
- Rights Agent GA — Gemini + Agent Stack
- Nanopayment royalty streams (per-play)
- ERC-721 receipt NFT per logged move
- Crew treasuries + on-chain cosigns

Street dance travels through the currencies that don't hold.

~450M

People aged 15–30 across the 15 shortlisted markets

15–20M

Estimated stablecoin users in the core four hubs today

~US\$80B

Annual remittance inflows into the shortlisted markets

Nigeria 33.7%

Philippines 3.8%

Argentina 211%

South Africa 5.3%

Turkey 69.8%

In these markets a card checkout costs 10–15% in FX and demands a card most young buyers don't hold. USDC on Arc settles in seconds, for sub-cent gas, with the creator's royalty split at the moment of sale.

BUILT FOR ENCODE CLUB · PROGRAMMABLE MONEY HACKATHON

Give the culture its receipts.

streetrail.lovable.app · [DanceMoveTokens.sol](#) on Arc Testnet

github.com/arunnadarasa/streetdancearc